Space Debris Detection Using Deep Learning and Exploratory Data Analysis

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***Abstract*—Space debris detection has become a critical challenge due to the increasing number of artificial objects orbiting Earth. This paper presents an exploratory data analysis (EDA) and a deep learning-based approach for detecting space debris using image datasets. The study analyzes dataset characteristics such as object size, distribution, and density, and proposes a YOLO-based object detection model for accurate and real-time debris detection. Experimental results demonstrate the effectiveness of the proposed method in identifying debris under challenging conditions.**

***Index Terms*—Space Debris, Object Detection, YOLO, CNN, Deep Learning, Computer Vision, EDA, mAP**

I. Introduction

The proliferation of satellite launches and space missions over the past two decades has resulted in a rapidly growing population of defunct spacecraft, rocket bodies, and collision fragments in low Earth orbit. As of 2024, the European Space Agency estimates over 36,500 trackable debris objects orbiting Earth, with millions of smaller fragments too small to track but large enough to cause catastrophic damage. Traveling at orbital velocities exceeding 7 km/s, even centimeter-scale debris carries sufficient kinetic energy to disable or destroy operational satellites. Traditional detection infrastructure — primarily ground-based radar and optical telescopes — struggles to track objects below 10 cm in size and is further constrained by weather, atmospheric interference, and limited coverage windows.

The urgency of this problem is compounded by the rapid commercialization of low Earth orbit, with mega-constellations such as Starlink and OneWeb deploying thousands of satellites and significantly increasing collision probability. In response, deep learning-based computer vision approaches have emerged as a compelling alternative to traditional detection pipelines. Convolutional neural networks and single-stage object detectors have demonstrated the ability to identify debris-like objects directly from telescope and camera imagery, offering potential for onboard real-time detection without dependence on ground infrastructure.

We present a **YOLOv9**-based detection framework for space debris identification in annotated satellite imagery. YOLOv9 introduces two key architectural innovations:

- **GELAN (Generalized Efficient Layer Aggregation Network)**— improves feature reuse across layers, which helps with small object detection since fine-grained spatial features aren't lost deep in the network.
- **PGI (Programmable Gradient Information)**— tackles the information bottleneck problem during training, meaning gradients carry richer signal through deep layers. This matters for low-contrast objects like debris on dark space backgrounds.

Prior to training, we perform systematic EDA to characterize object size distributions, spatial density, and bounding box geometry, using these findings to guide model configuration and preprocessing decisions..

The main contributions of this paper are as follows:

- A systematic EDA of the AIcrowd space debris dataset, revealing key statistical properties including bounding box size distribution, aspect ratio consistency, and the relationship between object density and scale
- Identification of core detection challenges — specifically the dominance of sub-20px debris objects and low signal-to-noise ratios — that inform model design decisions
- A YOLOv9-based detection pipeline trained and evaluated on approximately 27,000 annotated space images
- Quantitative evaluation using precision, recall, F1-score, and mAP@0.5, with analysis of failure cases on small object instances

The remainder of this paper is organized as follows: Section II presents the literature review, Section III describes the dataset, Section IV discusses the exploratory data analysis, Section V explains the proposed methodology, Section VI pro-

vides implementation details, Section VII presents results and evaluation, Section VIII shows visualization results, Section IX discusses findings, and Section X concludes the paper with future directions.

II. Literature Review

Space debris detection has been approached through three broad methodological categories: physics-based sensing methods (radar and optical), classical machine learning, and deep learning-based computer vision . Each paradigm offers distinct advantages but carries limitations that motivate the present work.

A. Traditional and Sensor-Based Detection Approaches

1) Radar-Based Detection (Muntoni et al. [1]): Radar-based methods provide real-time tracking with wide spatial coverage. However, performance degrades for objects < 10 cm due to limited resolution and background interference, making radar insufficient as a standalone solution in high-density LEO environments.

2) Optical and Space-Based Detection (Liu et al. [2]): Liu et al. propose a multi-star tracker network that repurposes satellite attitude sensors for debris detection. The system converts angular measurements into spatial coordinates using Gaussian MMSE correction, achieving positioning accuracy < 15 m and same-orbit recognition rates $> 80\%$. However, the approach is purely geometric and lacks visual feature learning, limiting its applicability to image-based pipelines. Additionally, optical systems are affected by atmospheric turbulence, lighting conditions, and pointing constraints.

3) Classical Machine Learning Approaches (Xi et al. [3]): Early approaches rely on feature extraction from candidate regions rather than end-to-end learning. Xi et al. use a neural network classifier on extracted debris regions, improving over traditional methods. However, reliance on hand-crafted proposals limits generalization across varying object scales and backgrounds. Classical methods such as SVMs and random forests also show limited scalability for large image datasets.

B. Deep Learning and YOLO-Based Approaches

Guo et al. [4] (Enhanced YOLOv8): Guo et al. introduce cross-scale feature fusion (CCFM) and CARAFE upsampling to improve detection of small, low-contrast debris. Their model achieves strong performance:

- mAP = 93.7%
- 35% reduction in parameters

Basit [5] (Adaptive YOLOv8): Basit applies an adaptive YOLOv8 framework for detecting 11 debris classes using SGD optimization and data augmentation (blur, grayscale). The model achieves mAP = 92.6%, but shows reduced performance:

- Small object precision drops to 0.65
- Degradation in high-density scenes

Roll et al. [6] (GraphRAG + LLaVA): Roll et al. propose a GraphRAG-based framework integrated with the LLaVA multimodal model. Their qualitative evaluation reports zero hallucinations across five test tasks, outperforming baseline and commercial models, although evaluation remains qualitative.

C. Summary and Research Gaps

Table I summarizes the key methods reviewed and their limitations. From the literature, it is observed that deep learning methods consistently outperform traditional sensor-based approaches in detection accuracy and scalability. However, several challenges remain unresolved:

- Detection of sub-pixel debris objects.
- Sensitivity to low signal-to-noise ratios.
- The lack of systematic data analysis to inform model design decisions.
- Limited real-time detection capability for onboard systems.

These gaps motivate the present work, in which we apply YOLOv9 — whose GELAN architecture preserves fine-grained spatial features and whose PGI mechanism reduces gradient information loss — to a 27,000-image annotated dataset, with model configuration directly informed by exploratory data analysis.

TABLE I
Comparison of Space Debris Detection Methods

Study	Method	Strength	Limitation
Muntoni et al. [1]	Radar	Real-time, wide coverage	Poor for <10 cm objects
Liu et al. [2]	Multi-star tracker	15 m positioning accuracy	No visual feature learning
Xi et al. [3]	Feature learning	Better than classical ML	Hand-crafted proposals
Guo et al. [4]	YOLOv8 + CCFM	93.7% mAP, lightweight	Tested on small dataset
Basit [5]	Adaptive YOLOv8	92.6% mAP, 11 classes	Struggles with small objects
Roll et al. [6]	GraphRAG + LLM	Zero hallucinations	Qualitative only
Ours	**YOLOv9 + EDA**	**EDA-informed design, PGI for small objects**	**Single dataset**

III. Dataset Description

The dataset used in this study is obtained from Kaggle and originates from the AIcrowd debris detection challenge . It consists of annotated space images containing debris objects with bounding box labels.

The dataset contains approximately 27,000 images divided into training, validation, and testing sets. Each image may contain one or multiple debris objects of varying sizes and shapes. The images are stored in JPEG format with a fixed resolution of 256×256 pixels.

Annotations are provided in CSV files, where bounding boxes are stored as lists in the format [xmin, xmax, ymin, ymax]. These values are initially stored as strings and require preprocessing to convert them into usable numerical formats.

TABLE II
DATASET SUMMARY

Property	Description
Source	Kaggle (AIcrowd Challenge)
Total Images	∼27,000
Training Set	20,000 images
Validation Set	2,000 images
Test Set	5,000 images
Image Size	256 × 256 pixels
Format	JPEG
Annotation Type	Bounding Boxes
Bounding Box Format	[xmin, xmax, ymin, ymax]



Fig. 1. Sample space image containing debris objects from the dataset.

The dataset presents several challenges for object detection tasks. Most debris objects are very small in size and appear in low-contrast backgrounds, making them difficult to detect. Ground-based telescope imagery of this kind typically exhibits low signal-to-noise ratios, as debris in orbit reflects limited sunlight and subtends a very small solid angle relative to the sensor. Additionally, images often contain multiple debris objects simultaneously, increasing scene-level complexity and placing greater demands on the model's localization capability. The variation in bounding box dimensions across the dataset further requires the detection model to maintain sensitivity across multiple scales. Collectively, these characteristics make the dataset a representative and challenging benchmark for evaluating deep learning models designed for small object detection in space imagery.

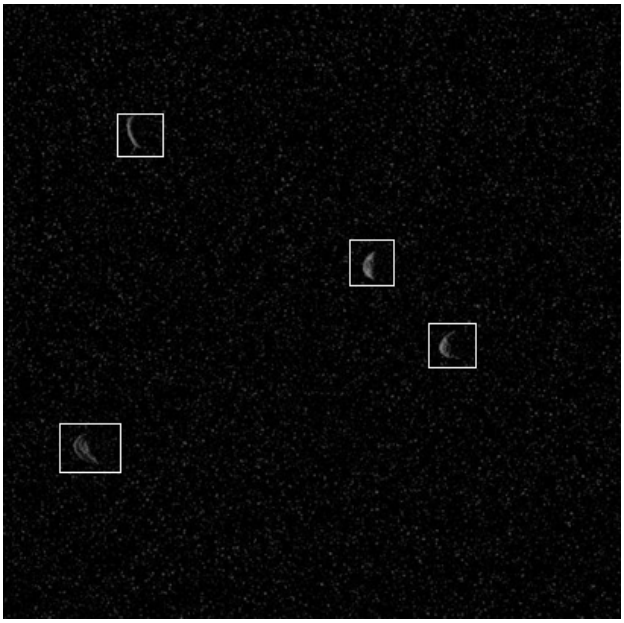


Fig. 2. Example image showing manual annotation for model training

IV. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the statistical properties of the dataset, including object size distribution, spatial density, and bounding box geometry. These insights directly inform model design choices for effective debris detection.

A. Bounding Box Size Distribution

The distributions of bounding box widths and heights (Fig. 3, Fig. 4) show a wide variation in object sizes, with a strong concentration of very small objects.

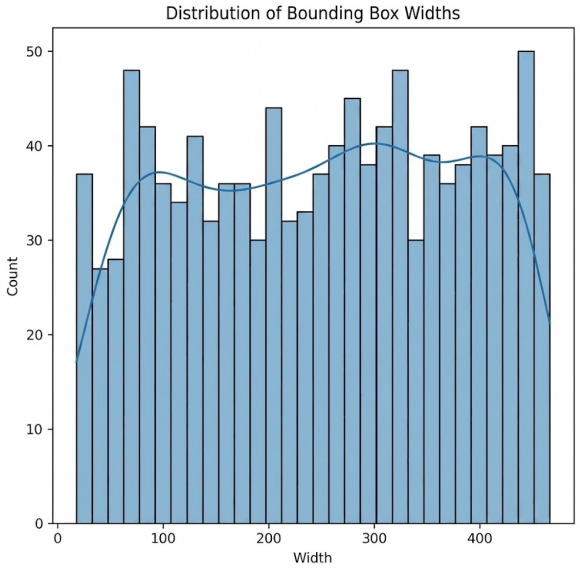


Fig. 3. Distribution of bounding box widths

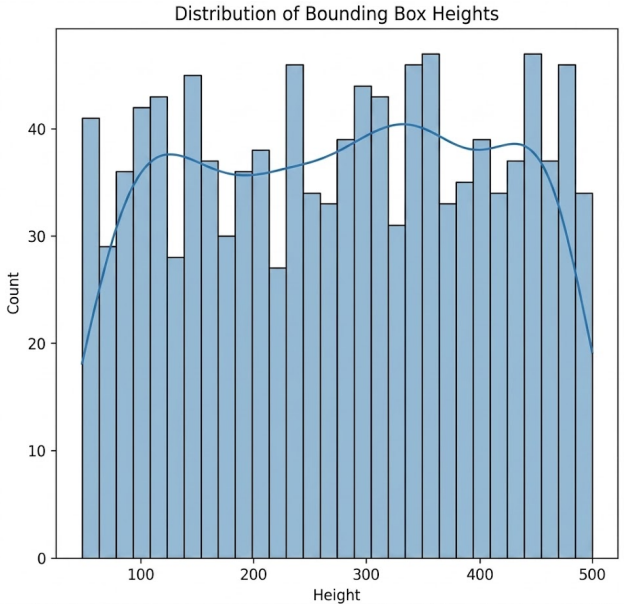


Fig. 4. Distribution of bounding box hieghts

Implication: The dominance of small objects necessitates the use of multi-scale feature extraction and high-resolution feature maps.

B. Number of Objects per Image

The distribution of object counts per image (Fig. 5) indicates that most images contain 3–5 debris objects, with some high-density cases.

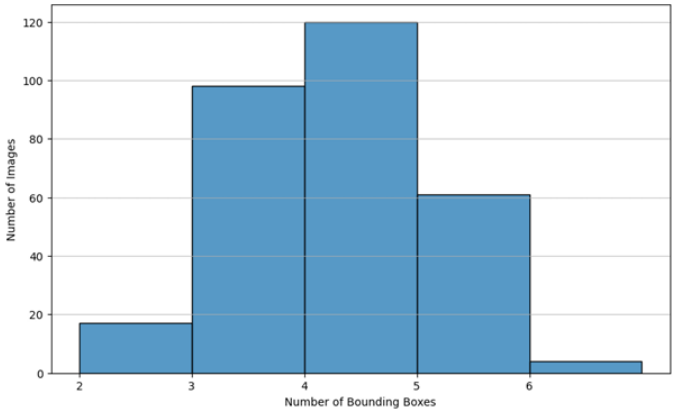


Fig. 5. Distribution of bounding box per image (first 300 image)

Implication: The model must handle multi-object detection efficiently while maintaining localization accuracy in dense scenes.

C. Aspect Ratio Distribution

Aspect ratios are concentrated within a narrow range (approximately 0.8–0.95), as shown in Fig. 6, indicating that most debris objects are nearly square in shape. . This observation implies that anchor box design can be simplified with limited aspect ratio variation, reducing the number of anchors required and improving overall detection efficiency. Aspect ratios are concentrated within a narrow range (approximately 0.8–0.95), as shown in Fig. 6, indicating that most debris objects are nearly square. This consistency suggests that debris fragments, regardless of their absolute size, tend to appear as compact, roughly isotropic targets within the image frame — likely a consequence of the point-like appearance of distant objects captured by wide-field telescopes.

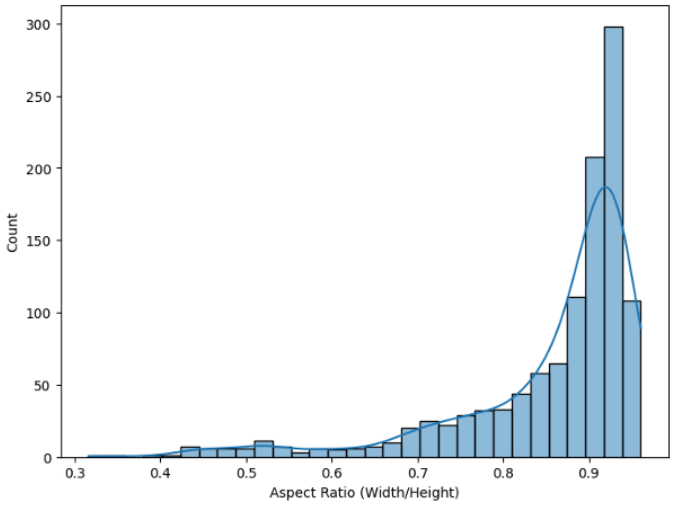


Fig. 6. Distribution of bounding box Aspect Ratios (Width/Hieght)

Implication: Anchor box design can be simplified with limited aspect ratio variation, improving detection efficiency.

D. Width–Height Relationship

A strong positive correlation between bounding box width and height is observed (Fig. 7), suggesting proportional scaling of object dimensions.

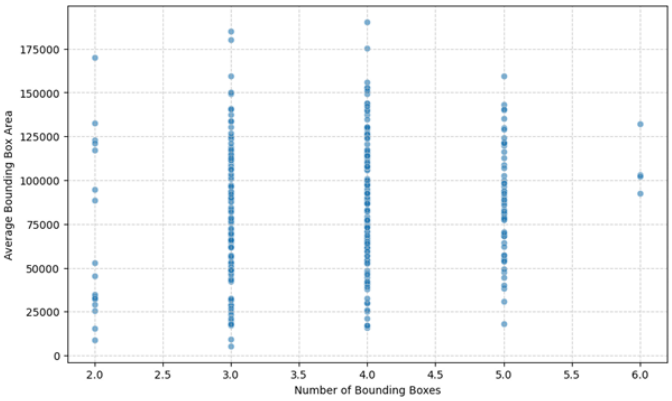


Fig. 7. Number of Bounding Boxes vs Average Bounding Box Area per Image

Implication: Consistent object geometry supports stable feature learning across scales.

E. Object Density vs Size

Analysis shows that images with higher object counts tend to contain smaller debris (Fig. 8).

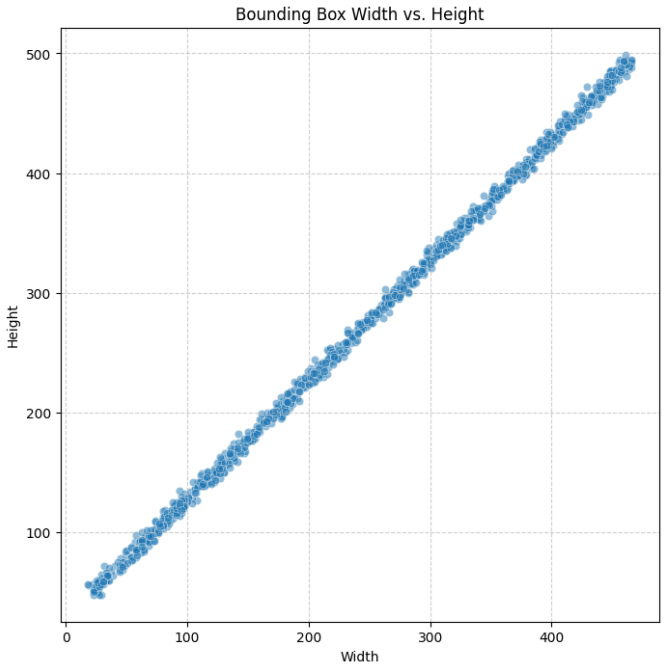


Fig. 8. Distibution Ratios

Implication: Dense scenes with small objects increase detection difficulty, requiring precise localization and robust feature representation.

F. Spatial Distribution of Objects

The distribution, as shown in Fig. 9, indicates that debris objects are uniformly distributed across the image space indicating no positional bias.

Unlike terrestrial object detection datasets that often exhibit center bias due to camera framing, space imagery captures debris in arbitrary positions relative to the sensor. This randomness increases the reliance on robust feature extraction rather than positional cues. As a result, the detection model must generalize across the entire image plane and remain sensitive to small, low-contrast objects irrespective of their location.

Additionally, the absence of spatial clustering suggests that preprocessing techniques such as tiling and dynamic cropping are beneficial, as they allow the model to examine localized regions at higher resolution without assuming any prior object placement. This observation reinforces the need for multi-scale feature learning and supports the architectural choice of YOLOv9 for handling dispersed small-object detection.

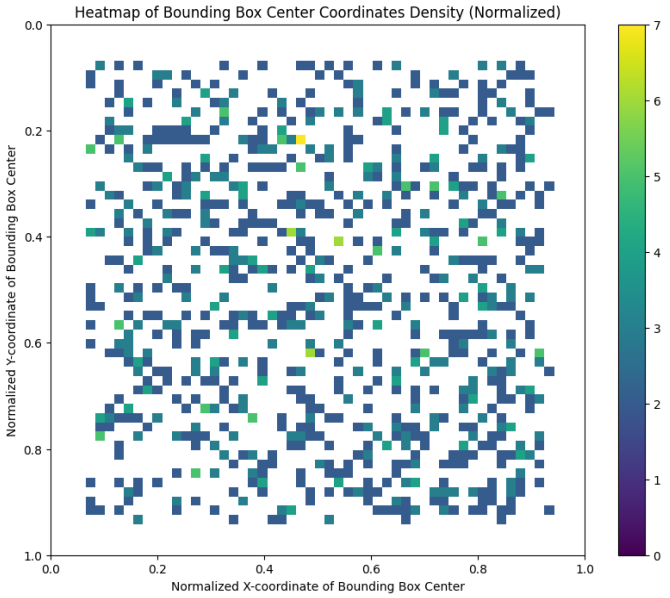


Fig. 9. Heatmap of Bounding Box (Spatial Awareness)

Implication: The model cannot rely on location priors hence detection must depend on visual features, not position. This increases detection difficulty and the model must generalize across the entire image space.

G. Key Insights

- A significant proportion of debris objects are extremely small in size, indicating a strong skew toward the lower end of the scale distribution
- Images typically contain 3–5 objects, reflecting moderate to high object density with occasional dense scenarios
- Substantial variation in object scale is observed, requiring robustness across multiple size ranges
- Low contrast between debris and background reduces visual separability and increases detection difficulty

H. Challenges

These observations highlight the inherent complexity of the dataset and directly translate into the following modeling challenges:

- Reliable detection of tiny objects remains difficult due to limited pixel representation
- Low signal-to-noise ratio adversely affects feature extraction and model sensitivity
- High object density introduces localization ambiguity and increases prediction complexity
- Imbalanced size distribution biases the model toward larger, more visible objects
- Real-time detection constraints impose additional computational and latency challenges

V. Proposed Methodology

The proposed system employs a deep learning-based object detection pipeline using a YOLOv9 architecture for space

debris identification. The workflow consists of data annotation, preprocessing, model training, and inference.

A subset of **754** images from the dataset was manually annotated using Roboflow to generate high-quality bounding box labels for debris objects. These annotations were used to create a structured training dataset suitable for supervised learning.

The annotated dataset was integrated with a larger corpus of approximately **27,000** images for training the YOLOv9 model. Preprocessing steps included image resizing, normalization, and conversion of annotation formats into YOLO-compatible labels.

The **YOLOv9** model was trained to learn spatial and contextual features for detecting debris objects of varying sizes, with particular emphasis on small object detection. During training, the model optimizes bounding box regression, objectness, and classification losses.

The overall methodology is designed to achieve accurate and efficient detection while maintaining real-time performance.

A. System Overview

The system follows a structured pipeline where input images are first preprocessed and annotated using Roboflow. The annotated dataset is then exported in YOLO format and used for training the YOLOv9 model on Google Colab with GPU acceleration.

During inference, the trained model processes input images in a single forward pass and predicts bounding boxes along with confidence scores for detected debris objects. The pipeline ensures efficient handling of large-scale datasets and supports real-time detection scenarios.

B. Data Preprocessing

The data preprocessing pipeline standardizes input images, enhances feature visibility, and improves model generalization. The key steps are as follows:

- **Dataset Acquisition and Annotation:**
  - Raw space debris images are collected from Kaggle.
  - A total of 754 images are manually annotated using Roboflow.
  - Bounding boxes are defined for debris objects using supervised annotation.
- **Preprocessing Operations (Roboflow Pipeline):**
  - Auto-orientation ensures consistent pixel alignment.
  - Grayscale conversion reduces computational complexity and emphasizes structural features.
  - Auto-contrast (contrast stretching) improves visibility in low-light conditions.
  - Unannotated images are filtered out to maintain dataset quality.
- **Image Resizing and Normalization:**
  - Images are resized to 640×640 pixels.
  - Pixel values are normalized for stable training.
- **Region Enhancement Techniques:**
  - Static cropping focuses on predefined regions of interest.
  - Dynamic cropping extracts annotated regions for better localization.
  - Tiling (2×2 split) improves detection of small debris objects.
- **Data Augmentation:**
  - Geometric transformations: rotation ($\pm 15°$, $90°$), shear ($\pm 10°$)
  - Photometric transformations: brightness ($\pm 15\%$), exposure ($\pm 10\%$), saturation ($\pm 25\%$), hue ($\pm 15°$)
  - Regularization techniques: noise injection, blur ($\leq$ 1.7 px), cutout
  - Dataset expanded to approximately 27,000 images
- **Data Formatting and Splitting:**
  - Dataset exported in YOLO format with normalized bounding boxes.
  - Split into training and validation sets for evaluation.

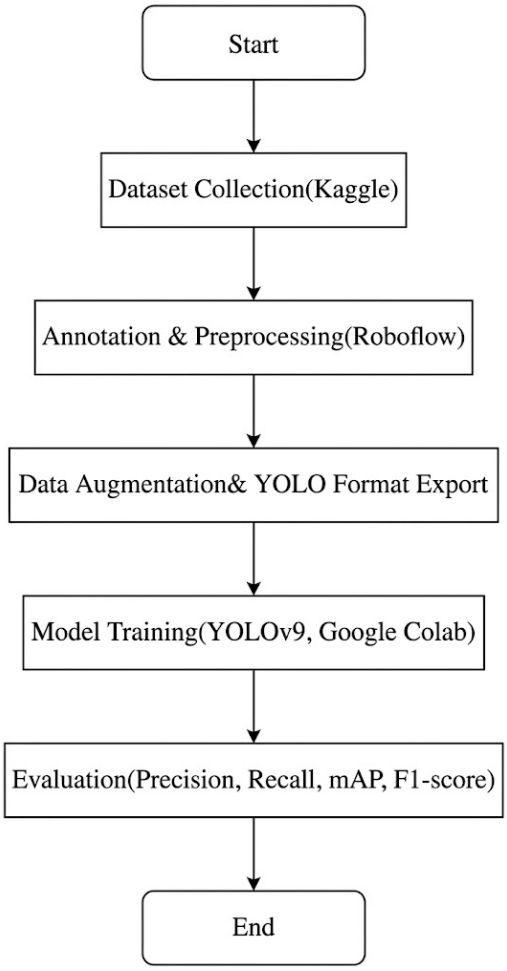


Fig. 10. Implementation workflow of debri detection system

C. Training Process

The model is trained using supervised learning on a dataset comprising manually annotated and augmented images. Train-

ing is conducted on Google Colab with GPU acceleration to efficiently handle the dataset scale.

The YOLOv9 model is trained for 50 epochs, optimizing a composite loss function consisting of:

- Bounding box regression loss for accurate localization
- Classification loss for correct object identification
- Distribution Focal Loss (DFL) for improved bounding box precision

Hyperparameters, including learning rate and batch size, are tuned to ensure stable convergence. Model parameters are iteratively updated via backpropagation to minimize the overall loss and improve detection performance across epochs.

D. Prediction and Post-processing

During inference, the trained YOLOv9 model processes input images in a single forward pass and outputs bounding boxes with corresponding confidence scores.

To refine predictions, Non-Maximum Suppression (NMS) is applied to remove redundant overlapping detections and retain the most confident bounding boxes. The final output consists of accurately localized debris objects with associated confidence values, enabling reliable detection in complex scenarios.

VI. Results and Evaluation

The performance of the proposed model is evaluated using standard object detection metrics, including precision, recall, F1-score, and mean Average Precision (mAP). The training and validation curves demonstrate stable convergence behavior, as shown in Fig. 11 and Fig. 12.

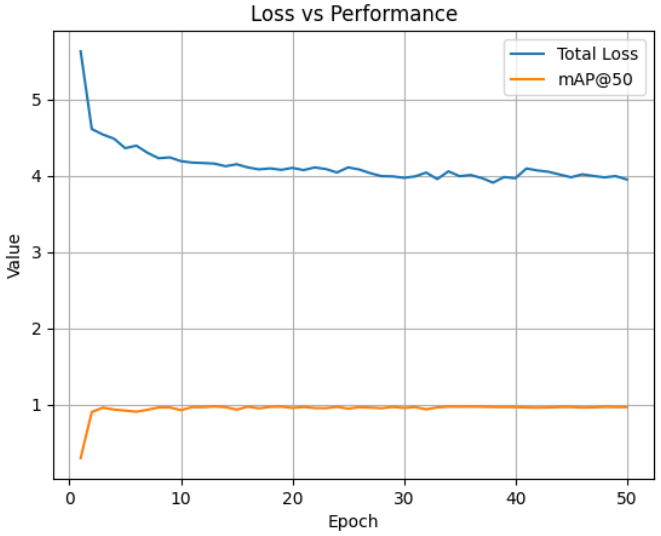


Fig. 11. Loss vs Performance over epochs

In Fig. 10. training loss decreases sharply during early epochs and stabilizes after convergence, indicating effective feature learning. The minimal gap between training and validation loss suggests good generalization with no significant overfitting.

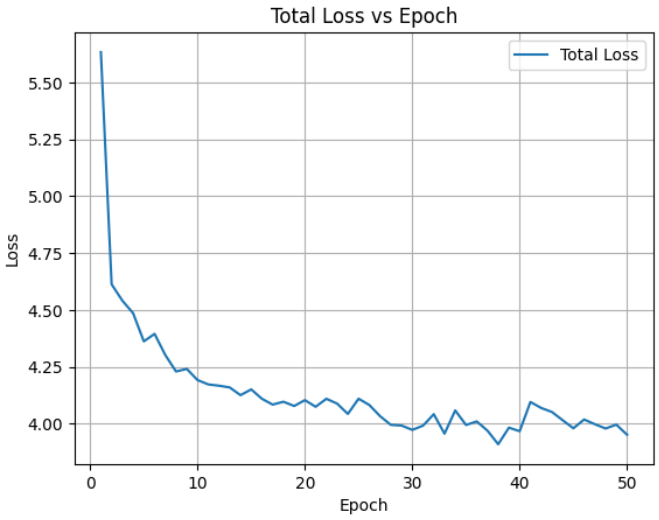


Fig. 12. Total Loss vs Epoch

Precision measures how many predicted bounding boxes correctly match debris objects, while recall measures how many actual debris objects the model detects. Together, they capture complementary aspects of performance.

- **High Precision** indicates few false positives — predicted debris objects are usually correct.
- **Lower Recall** shows the model misses some objects, especially small or low-contrast ones below the detection threshold.

Both precision and recall stabilize around 0.94–0.96 (Fig. 12), showing a strong balance between detection accuracy and false positives.

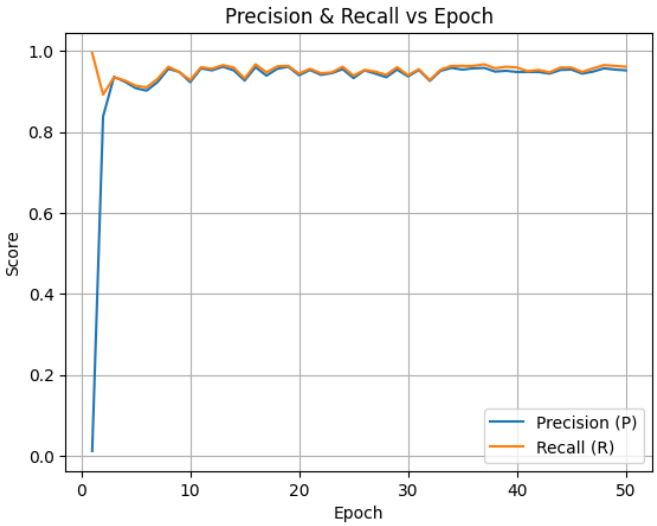


Fig. 13. Precision and Recall vs Epoch

While precision remains high, recall is comparatively lower, indicating missed detections—particularly for small and low-contrast debris identified during EDA.

The F1-score is defined as the harmonic mean of precision and recall. As shown in Fig. 14, the F1-score remains consistently high (approximately 0.95) across training epochs, indicating that the model maintains a strong balance between precision and recall.

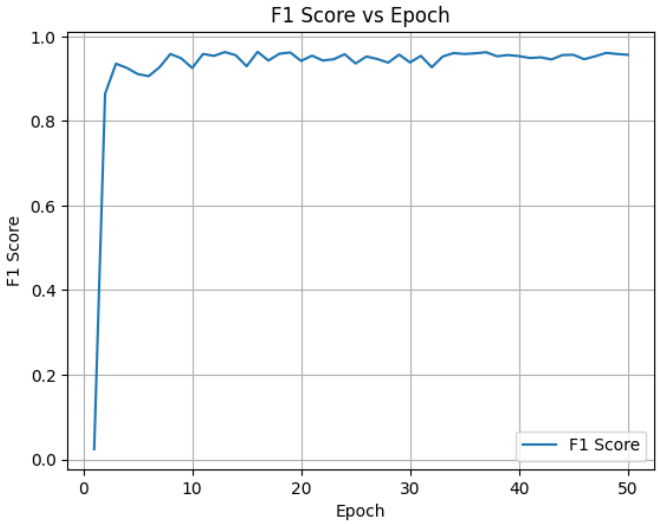


Fig. 14. F1 Score vs Epoch

The model achieves a high mAP@50 (0.97), while mAP@50–95 remains around 0.43, as depicted in Fig. 15. This indicates strong detection capability but relatively lower localization precision under stricter IoU thresholds.

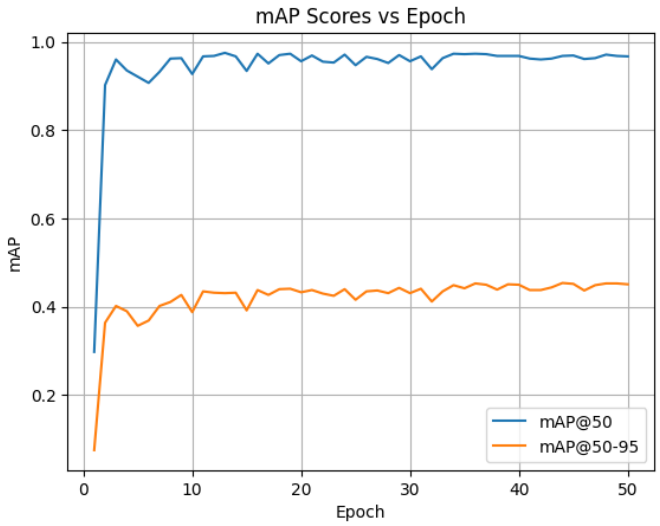


Fig. 15. mAP Scores vs Epoch

Training loss components, including box loss, classification loss, and DFL loss, are shown in Fig. 16. Classification loss decreases significantly, while box and DFL losses stabilize over time, indicating improved bounding box regression.

Overall, the model demonstrates strong performance in detecting medium and large debris objects. However, detection performance for very small objects remains limited due to localization challenges.



Fig. 16. Training Loss Components

TABLE III
COMPARISON OF MODEL PERFORMANCE ACROSS VERSIONS

Metric	Version 2 (YOLOv9)	Version 1 (YOLOv5)
Accuracy	∼0.95	∼0.90
Precision	0.95	0.895
Recall	0.94	0.87
F1-Score	0.95	0.91
mAP@50	0.97	0.95
mAP@50–95	0.43	—

VII. CONCLUSION

This study presents an EDA-driven deep learning framework for space debris detection using the YOLOv9 object detection model. The work integrates systematic exploratory data analysis with a robust detection pipeline to address challenges associated with small object detection in space imagery.

The EDA reveals critical dataset characteristics, including the dominance of very small debris objects, moderate-to-high object density (typically 3–5 objects per image), and low contrast between objects and background. These insights directly inform preprocessing strategies such as tiling, augmentation, and resolution scaling to improve detection performance.

The proposed YOLOv9-based model, trained on an augmented dataset of approximately 27,000 images, demonstrates strong performance with precision (∼0.95), recall (∼0.94), F1-score (∼0.95), and mAP@50 (∼0.97). Training results show stable convergence, with decreasing loss values and consistent improvement across epochs.

A comparative analysis with a baseline YOLOv5 model (taken form model made on ROBOFLOW) highlights significant improvements in detection accuracy and overall performance, confirming the effectiveness of the proposed approach. Despite these advancements, the model still faces challenges in detecting extremely small debris objects due to limited pixel representation and localization constraints.

Overall, the integration of EDA with deep learning provides a structured and effective solution for space debris detection. The proposed methodology is scalable and suitable for real-time applications, with potential for further enhancement through improved small-object detection techniques and more diverse datasets.

VIII. Future Work

Future work includes improving detection accuracy for small debris, applying advanced models such as transformer-based architectures, and integrating real-time detection systems.

Guidance

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